

FinalProject - Mathew Scaria

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Introduction

Research Question: How do Vacancy Rates Affect Fair Market Rent? Motivation: This is an important research topic because vacancy rates and fair market rent are closely tied to housing affordability, urban development, and economic stability. Understanding this relationship can help policymakers, landlords, and renters better understand how housing markets function.

Variables:

Dependent Variable: `fmr_2bed` (Rent for a 2 bedroom rental unit) Independent Variable: `vacancy_rate` (The percent of rental units that are vacant for rent) Controls: `total_housing_units` (Total number of rental units this includes apartments, houses, etc) `median_income` (People's Income in dollars) `households_with_children` (The number of families that have children under 18)

1. Importing my Data

```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'  
  
## The following objects are masked from 'package:stats':  
##  
##   filter, lag  
  
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union
```

```
library(stargazer)
```

```
##  
## Please cite as:  
  
## Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics Tables.  
  
## R package version 5.2.3. https://CRAN.R-project.org/package=stargazer
```

```
FinalData <- read.csv("FinalProjectdataset.csv") #giving my data a name
```

```
head(FinalData) #This lets me see the first couple rows of data
```

```
##   fips_code   county_name state fmr_2bed vacancy_rate total_housing_units
## 1    1001 Autauga County    AL    977      8.928066         24731
## 2    1003 Baldwin County    AL   1206     26.358954        128518
## 3    1005 Barbour County    AL    740     22.406426         11702
## 4    1007  Bibb County     AL   1075     16.710671          9090
## 5    1009 Blount County     AL   1075     11.358045         24793
## 6    1011 Bullock County    AL    820     24.176548          4554
##   median_income households_with_children
## 1         69841                5155
## 2         75019                16692
## 3         44290                 962
## 4         51215                1187
## 5         61096                4405
## 6         36723                 402
```

```
# Summary statistics for the variables I will use in my regressions
```

```
summary(select(FinalData, c('fmr_2bed', 'vacancy_rate', 'households_with_children', 'median_income', 'total_housing_units')))
```

```
##      fmr_2bed      vacancy_rate      households_with_children median_income
## Min.   : 444.0   Min.   : 1.707   Min.   : 0.0           Min.   : 16170
## 1st Qu.: 795.0   1st Qu.: 9.514   1st Qu.: 652.5         1st Qu.: 54097
## Median : 858.0   Median :15.011   Median : 1583.0        Median : 63128
## Mean   : 965.3   Mean   :17.052   Mean   : 7157.6         Mean   : 64967
## 3rd Qu.:1024.5   3rd Qu.:21.603   3rd Qu.: 4475.0        3rd Qu.: 73083
## Max.   :3293.0   Max.   :77.316   Max.   :594569.0       Max.   :178707
## total_housing_units
## Min.   : 47
## 1st Qu.: 5340
## Median : 12375
## Mean   : 44339
## 3rd Qu.: 30857
## Max.   :3624084
```

2. OLS Regression

This is my baseline OLS regression using all four variables on the full county-level dataset. I regress 2-bedroom Fair Market Rent on vacancy rate, total vacant units, and median income.

```
# OLS regression on the full dataset
```

```
reg_full <- lm(fmr_2bed ~ vacancy_rate + total_housing_units + median_income + households_with_children)
summary(reg_full)
```

```
##
## Call:
## lm(formula = fmr_2bed ~ vacancy_rate + total_housing_units +
```

```
##      median_income + households_with_children, data = FinalData)
##
## Residuals:
##      Min        1Q    Median        3Q        Max
## -1404.17  -104.42   -26.70    69.14   1803.41
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      1.935e+02  1.611e+01  12.009 < 2e-16 ***
## vacancy_rate      1.323e+00  3.427e-01   3.862 0.000115 ***
## total_housing_units 1.218e-03  1.182e-04  10.310 < 2e-16 ***
## median_income      1.106e-02  2.046e-04  54.069 < 2e-16 ***
## households_with_children -3.302e-03  7.066e-04  -4.672 3.1e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 186.8 on 3206 degrees of freedom
## Multiple R-squared:  0.6122, Adjusted R-squared:  0.6117
## F-statistic: 1265 on 4 and 3206 DF,  p-value: < 2.2e-16
```

3. Filtered Regression (High Income Counties)

Here I filter the dataset to only include counties where median household income is above \$100,000. I want to see whether the relationship between vacancy rates and rent looks different in wealthier counties compared to the full sample.

```
# Filter to high income counties only (median income above $90,000)
FinalData_high_income <- filter(FinalData, median_income > 90000)

cat("Number of counties in filtered dataset:", nrow(FinalData_high_income), "\n")
```

```
## Number of counties in filtered dataset: 248
```

```
# OLS regression on the filtered high income dataset
reg_filtered <- lm(fmr_2bed ~ vacancy_rate + total_housing_units + median_income + households_with_children, data = FinalData_high_income)
summary(reg_filtered)
```

```
##
## Call:
## lm(formula = fmr_2bed ~ vacancy_rate + total_housing_units +
##      median_income + households_with_children, data = FinalData_high_income)
##
## Residuals:
##      Min        1Q    Median        3Q        Max
## -807.45  -255.50   -34.84   162.45  1676.58
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      8.067e+01  1.778e+02   0.454 0.650454
## vacancy_rate      4.872e+00  2.192e+00   2.223 0.027147 *
```

```
## total_housing_units      2.735e-03  4.463e-04   6.128 3.57e-09 ***
## median_income            1.234e-02  1.644e-03   7.508 1.14e-12 ***
## households_with_children -8.938e-03  2.308e-03  -3.873 0.000138 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 372 on 243 degrees of freedom
## Multiple R-squared:  0.3752, Adjusted R-squared:  0.3649
## F-statistic: 36.47 on 4 and 243 DF,  p-value: < 2.2e-16
```

4. Preferred Regression (Log-Transformed Income)

My preferred regression uses the full dataset but log-transforms median income. Income tends to have a non-linear relationship with rent — an extra dollar of income matters more at lower income levels than at higher ones. Log-transforming income captures this and typically improves model fit.

```
# Create a log income variable
FinalData <- mutate(FinalData, log_income = log(median_income))

# Preferred regression with log-transformed income
reg_preferred <- lm(fmr_2bed ~ vacancy_rate + total_housing_units + log_income + households_with_children, data = FinalData)
summary(reg_preferred)
```

```
##
## Call:
## lm(formula = fmr_2bed ~ vacancy_rate + total_housing_units +
##     log_income + households_with_children, data = FinalData)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1667.65  -119.49   -41.75    72.95   1950.42
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -5.721e+03  1.504e+02  -38.039  < 2e-16 ***
## vacancy_rate    8.570e-01  3.724e-01   2.301   0.0214 *
## total_housing_units  6.650e-04  1.266e-04   5.253   1.6e-07 ***
## log_income      6.012e+02  1.347e+01  44.631  < 2e-16 ***
## households_with_children  5.922e-04  7.520e-04   0.787   0.4311
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 202.8 on 3206 degrees of freedom
## Multiple R-squared:  0.5427, Adjusted R-squared:  0.5421
## F-statistic: 951.2 on 4 and 3206 DF,  p-value: < 2.2e-16
```

5. Stargazer Table

The table below displays all three regressions side by side for easy comparison.

*#This table lets me create a singular table that lets me see all three of my regressions
#I used claude for this section to put the stargazer table together*

```
stargazer(reg_full, reg_filtered, reg_preferred,
          type           = "latex",
          title           = "OLS Regression Results: Fair Market Rent (2-Bedroom)",
          dep.var.labels = "2-Bedroom Fair Market Rent (USD)",
          covariate.labels = c("Vacancy Rate (%)",
                               "Total Vacant Units",
                               "Median Income (USD)",
                               "Log Median Income"),
          column.labels   = c("Full Sample", "High Income Counties", "Preferred (Log Income)"),
          omit.stat        = c("f", "ser"),
          digits           = 3,
          notes            = "Data: HUD Fair Market Rent API and U.S. Census Bureau ACS 2023.",
          notes.align      = "l")
```

% Table created by stargazer v.5.2.3 by Marek Hlavac, Social Policy Institute. E-mail: marek.hlavac at gmail.com % Date and time: Tue, May 12, 2026 - 10:41:30 AM

```
library(tinytex)
```

```
## Warning: package 'tinytex' was built under R version 4.5.3
```

Table 1: OLS Regression Results: Fair Market Rent (2-Bedroom)

	<i>Dependent variable:</i>		
	2-Bedroom Fair Market Rent (USD)		
	Full Sample	High Income Counties	Preferred (Log Income)
	(1)	(2)	(3)
Vacancy Rate (	(0.343)	(2.192)	(0.372)
Total Vacant Units	0.001*** (0.0001)	0.003*** (0.0004)	0.001*** (0.0001)
Median Income (USD)	0.011*** (0.0002)	0.012*** (0.002)	
Log Median Income			601.187*** (13.470)
households_with_children	−0.003*** (0.001)	−0.009*** (0.002)	0.001 (0.001)
Constant	193.520*** (16.115)	80.672 (177.812)	−5,720.726*** (150.392)
Observations	3,211	248	3,211
R ²	0.612	0.375	0.543
Adjusted R ²	0.612	0.365	0.542

Note:

*p<0.1; **p<0.05; ***p<0.01

Data: HUD Fair Market Rent API and U.S. Census Bureau ACS 2023.